- GEMODELTOOLS -

HANC MODEL WITH HETEROGENEOUS RETURNS

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Model

We consider a *closed* economy with heterogeneous agents and *flexible prices and wages*. Relative to the standard HANC model, households now differ in the return they earn on their wealth.

Time is discrete and indexed by t. There is a continuum of households indexed i.

Households. Households are heterogeneous *ex post* with respect to their labor productivity, e_{it}, their excess return, r_{it}^X, and their assets, a_{it-1}. All households share the discount factor β. Each period households exogenously supply $\ell_{it} = e_{it}$ units of labor, earn the individual return $r_t + r_{it}^X$ on the assets they bring into the period, and choose consumption c_{it} subject to a no-borrowing constraint, $a_{it} \geq 0$. Households have *perfect foresight* wrt. the interest rate and the wage rate, $\{r_t, w_t\}_{t=0}^{\infty}$, and solve the problem

$$
\begin{aligned}
v_t(e_{it}, r_{it}^X, a_{it-1}) = \max_{a_{it}, c_{it}} \frac{c_{it}^{1-\sigma}}{1-\sigma} + \beta \mathbb{E}_t \left[v_{t+1}(e_{it+1}, r_{it+1}^X, a_{it}) \right] \qquad (1) \\
\text{s.t.} \\
a_{it} + c_{it} = (1 + r_t + r_{it}^X) a_{it-1} + w_t e_{it} \\
\log e_{it} = \rho_e \log e_{it-1} + \psi_{it}, \;\; \psi_{it} \sim \mathcal{N}(\mu_\psi, \sigma_\psi^2), \;\; \mathbb{E}[e_{it}] = 1 \\
r_{it}^X = (1-\rho_r)\bar{r}^X + \rho_r r_{it-1}^X + \epsilon_{it}, \;\; \epsilon_{it} \sim \mathcal{N}(0, \sigma_r^2) \\
a_{it} \geq 0,
\end{aligned}
$$

where ψ_{it} and ϵ_{it} are independent, and implicitly

$$
v_t(e_{it}, r_{it}^X, a_{it-1}) = v(e_{it}, r_{it}^X, a_{it-1}, \{r_\tau, w_\tau\}_{\tau=t}^{\infty}).
$$

The excess return r_{it}^X is known at the beginning of period t, and has the unconditional mean $\bar{r}^X$.

We denote optimal policy functions by $a_t^*(e_{it}, r_{it}^X, a_{it-1})$ and $c_t^*(e_{it}, r_{it}^X, a_{it-1})$. The distribution of households *before* the realization of idiosyncratic shocks, i.e. over e_{it-1}, r_{it-1}^X and a_{it-1}, is denoted $\underline{\boldsymbol{D}}_t$. The distribution of households *after* the realization of idiosyncratic shocks, i.e. over e_{it}, r_{it}^X and a_{it-1}, is denoted $\boldsymbol{D}_t$.

Central aggregate variables are

$$A_t^{hh} = \int a_t^*(e_{it}, r_{it}^X, a_{it-1}) d\boldsymbol{D}_t = \boldsymbol{a}_t^{*\prime} \boldsymbol{D}_t \tag{2}$$

$$C_t^{hh} = \int c_t^*(e_{it}, r_{it}^X, a_{it-1}) d\boldsymbol{D}_t = \boldsymbol{c}_t^{*\prime} \boldsymbol{D}_t \tag{3}$$

$$RETURNS_t^{hh} = \int (r_t + r_{it}^X) a_{it-1} d\boldsymbol{D}_t \tag{4}$$

$$L_t^{hh} = \int e_{it} d\boldsymbol{D}_t = 1. \tag{5}$$

$RETURNS_t^{hh}$ is the total capital income received by households. In the standard HANC model it equals $r_t A_{t-1}^{hh}$. Here it does not equal $(r_t + \bar{r}^X) A_{t-1}^{hh}$ in general, because households with a high r_{it}^X accumulate more wealth.

To solve the model, we define the beginning-of-period value function as

$$\underline{v}_t \left(e_{it-1}, r_{it-1}^X, a_{it-1} \right) = \mathbb{E}_t \left[v_t(e_{it}, r_{it}^X, a_{it-1}) \right]. \tag{6}$$

The *envelope condition* implies

$$\underline{v}_{t,a}(e_{it-1}, r_{it-1}^X, a_{it-1}) = \frac{\partial \underline{v}_t(e_{it-1}, r_{it-1}^X, a_{it-1})}{\partial a_{it-1}} = \mathbb{E}_t \left[(1 + r_t + r_{it}^X) c_{it}^{-\sigma} \right]. \tag{7}$$

Note that the return r_{it}^X sits *inside* the expectation. The *first order condition* for consumption implies

$$c_{it}^{-\sigma} = \beta \underline{v}_{t+1,a}(e_{it}, r_{it}^X, a_{it}). \tag{8}$$

Firms. A representative firm rents capital, K_{t-1}, and hires labor, L_t, to produce goods, with the production function

$$Y_t = \Gamma_t K_{t-1}^{\alpha} L_t^{1-\alpha}, \;\; \alpha \in (0,1), \tag{9}$$

where Γ_t is technology. Capital depreciates with the rate δ. Profit maximization

$$\max_{K_{t-1}, L_t} Y_t - w_t L_t - r_t^k K_{t-1}$$

implies the standard pricing equations

$$r_t^k = \alpha \Gamma_t (K_{t-1}/L_t)^{\alpha-1} \tag{10}$$

$$w_t = (1-\alpha)\Gamma_t (K_{t-1}/L_t)^{\alpha}, \tag{11}$$

where r_t^k is the rental rate of capital and w_t is the wage rate.

Mutual fund. A zero-profit mutual fund owns all the capital. It takes deposits from households, $A_t = K_t$, earns $r_t = r_t^k - \delta$ per unit of capital, and pays household i the return $r_t + r_{it}^X$. Zero profit requires that the fund pays out exactly what it earns,

$$RETURNS_t^{hh} = r_t K_{t-1}. \tag{12}$$

Given $A_{t-1}^{hh} = K_{t-1}$, this is the same as $\int r_{it}^X a_{it-1} d\boldsymbol{D}_t = 0$: the excess returns only redistribute capital income across households, and the *wealth-weighted* average return equals r_t.

Market clearing. Market clearing requires

$$\text{Capital: } K_t = A_t = A_t^{hh} \tag{13}$$

$$\text{Labour: } L_t = L_t^{hh} = 1 \tag{14}$$

$$\text{Goods: } Y_t = C_t^{hh} + \underbrace{K_t - K_{t-1} + \delta K_{t-1}}_{=I_t}, \tag{15}$$

where I_t is investment.

Steady state and calibration

We only study the steady state. We normalize $Y = 1$ and $L = 1$, and target $K/Y = 3$ and $r = 0.04$. This gives directly

$$K = 3, \;\; \Gamma = \frac{Y}{K^{\alpha} L^{1-\alpha}}, \;\; r^k = \alpha \Gamma (K/L)^{\alpha-1}, \;\; \delta = r^k - r, \;\; w = (1-\alpha)\Gamma (K/L)^{\alpha}.$$

Given r and w, we solve for the two unknowns β and $\bar{r}^X$ such that

$$A^{hh} - K = 0 \;\; \text{(capital market clearing)}$$
$$RETURNS^{hh} - rK = 0 \;\; \text{(the fund makes zero profit).}$$

Because wealth is concentrated among households with high r_{it}^X, the calibrated $\bar{r}^X$ is negative.

The remaining parameters are

Parameter	Description	Value
σ	CRRA coefficient	2.0
α	Capital share	0.36
ρ_e	Persistence of $\log e_{it}$	0.95
σ_ψ	Std. of ψ_{it}	$0.30\sqrt{1-\rho_e^2}$
ρ_r	Persistence of r_{it}^X	0.8
σ_r	Std. of ϵ_{it}	0.019
N_e	Grid points for e_{it}	5
N_r	Grid points for r_{it}^X	7

As a benchmark, we also solve the model without return heterogeneity, $\sigma_r = 0$. The calibration then gives $\bar{r}^X = 0$.

Implementation

The **files** are: `blocks.py`, `household_problem.py`, `steady_state.py`, and `HANCModel.py`.

The results are produced in the **notebook** `HANC.ipynb`.

Two exogenous states. `GEModelTools` only has one stochastic discrete state, z. We stack r^X and e into one state with $N_z = N_r N_e$ points. With Π_r and Π_e the two transition matrices, the joint transition matrix and ergodic distribution are the Kronecker products

$$\Pi_z = \Pi_r \otimes \Pi_e, \quad \pi_z = \pi_r \otimes \pi_e.$$

With r^X first, the state `i_z` maps to `i_r = i_z // Ne` and `i_e = i_z % Ne`.

Step 1. In the `.settings` method the household block now also returns the capital income of each household:

```
self.grids_hh = ['a'] # grids
self.pols_hh = ['a'] # policy functions
self.inputs_hh = ['r','w'] # direct inputs
self.inputs_hh_z = [] # transition matrix inputs
self.outputs_hh = ['a','c','returns'] # outputs
self.intertemps_hh = ['vbeg_a'] # intertemporal variables
```

Step 2. In the `.setup` method the sizes of the exogenous grids are:

```
par.Nfix = 1 # number of fixed states
par.Nr = 7 # number of points in r grid
par.Ne = 5 # number of productivity states
par.Nz = par.Ne*par.Nr # number of stochastic discrete states
```

Step 3. In the `steady_state.py` file the function `prepare_hh_ss` sets the two grids and combines them:

```
par.e_grid[:],e_trans,e_ergodic,_,_ = log_rouwenhorst(par.rho_e,par.
    sigma_psi,par.Ne)
par.r_grid[:],r_trans,r_ergodic,_,_ = rouwenhorst(par.rX_mean*(1-par.
    rho_r),par.rho_r,par.sigma_r,par.Nr)
ss.z_trans[0,:,:] = np.kron(r_trans,e_trans) # shape=(Nz,Nz)
ss.Dbeg[0,:,0] = np.kron(r_ergodic,e_ergodic) # all start at a_lag = 0
```

Step 4. In the `household_problem.py` file the function `solve_hh_backwards` uses the individual return in the cash-on-hand, in the capital income and in the envelope condition:

```
for i_z in range(par.Nz):
    i_r = i_z // par.Ne
    i_e = i_z % par.Ne
    r_eff = r + par.r_grid[i_r]
    ... # EGM step as in the standard model, with r_eff and e_grid[i_e]
    returns[0,i_z] = r_eff*par.a_grid
    v_a[i_z] = (1+r_eff)*c[0,i_z]**(-par.sigma)
vbeg_a[0] = z_trans[0]@v_a
```

- **Note:** v_a must be computed for all i_z *before* taking the expectation, since $1 + r_t + r_{it}^X$ differs across states.
- **Check:** Solve the household problem in partial equilibrium:

```
model.ss.r = 0.04
model.ss.w = 0.64
model.par.beta = 0.89
model.par.rX_mean = 0.0
model.solve_hh_ss(do_print=True) # calls prepare_hh_ss
model.simulate_hh_ss(do_print=True)
model.ss.RETURNS_hh/model.ss.A_hh # average return
```

Step 5. In the `steady_state.py` file the function `find_ss` sets the targets and closed forms, then solves for β and $\bar{r}^X$ with `scipy.optimize.root`, using the two conditions above:

```
def obj_ss(x,model):
    par.beta = x[0]
    par.rX_mean = x[1]
    model.solve_hh_ss() # calls prepare_hh_ss, which rebuilds the r grid
    model.simulate_hh_ss()
    ss.clearing_A = ss.A-ss.A_hh
    clearing_returns = ss.RETURNS_hh-ss.r*ss.K
    return np.array([ss.clearing_A,clearing_returns])
```